

Kill-Test v6 Universality: EFC vs NFW across 175 SPARC Galaxies

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Abstract. Kill-Test v6 probe-2 reported EFC superiority over NFW on 5 SPARC galaxies. This note extends probe-2 to all 175 galaxies in the SPARC database (Lelli, McGaugh & Schombert 2016), fitting both the EFC phenomenological rotation curve model (3 parameters) and the NFW dark matter halo profile (2 parameters) to each galaxy independently using differential evolution. Of 171 galaxies with sufficient data (≥ 5 points), 103 (60.2%) favour EFC by AIC, with median $\Delta\text{AIC} = +6.21$. The regime classification (FLOW/TRANSITION/LATENT) shows statistically significant separation (Mann-Whitney $p \approx 0$), with no significant mass dependence (Spearman $\rho = 0.11$, $p = 0.15$). The cherry-picking objection is refuted. This is a *model compression test*, not a theory test: the result establishes that a specific low-dimensional functional form — motivated by EFC entropy-gradient phenomenology — outperforms the standard halo profile under equal information-theoretic penalties. It does not, by itself, validate EFC as a gravitational theory. Key limitations include the absence of baryonic coupling in the NFW comparison, the use of point estimates rather than full posteriors, and the post-hoc nature of the LATENT-galaxy interpretation. Concrete upgrades (NFW+baryons, nested sampling, predictive LATENT criterion) are specified.

Keywords: Energy-Flow Cosmology, rotation curves, SPARC, Kill-Test, NFW, model comparison, AIC, regime classification

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1. Introduction

Kill-Test v6 [1] established a six-probe comparison between Energy-Flow Cosmology (EFC) and Λ CDM across regimes L0–L3. Probe-2 (“Multi-component SPARC refit”) tested the EFC phenomenological rotation curve model against NFW on 5 galaxies, reporting 100% EFC success with ΔAIC values reaching -126 (NGC 7331).

The natural counter-argument is cherry-picking: 5 out of 175 galaxies could be a biased sample. This note eliminates that objection by extending probe-2 to the complete SPARC Rotmod database [2], fitting all 175 galaxies with identical methodology.

1.1. Scope and limitations

This is a *phenomenological* comparison in the L3 (galactic) regime. The EFC model $v(r) = v_{\text{flat}} \sqrt{1 - \exp(-(r/r_{\text{turnon}})^{\text{sharpness}})}$ is a 3-parameter curve that captures the entropy-gradient phenomenology of flat rotation curves. It is *not* derived from the EFC action; it is the empirical fitting function calibrated in the original SPARC 175 analysis [3].

The NFW profile $v^2(r) = V_{200}^2 \cdot [\ln(1+x) - x/(1+x)] / [\ln(1+c) - c/(1+c)]$ is the standard 2-parameter dark matter halo model. The comparison is therefore EFC-phenomenological (3 params) vs CDM-halo (2 params), with AIC penalising EFC by +2 for the extra parameter.

This analysis is independent of:

- The Background No-Go Theorem [4], which concerns $H(z)$ modifications
- Scale-Localised Modified Gravity [5], which concerns perturbation-sector $E_G(k, z)$ predictions at L1–L2 scales

2. Method

2.1. Data

The SPARC Rotmod database [2] provides observed rotation curves for 175 late-type galaxies with radii, observed velocities, velocity uncertainties, and baryonic mass decomposition. We use the observed velocity V_{obs} and its uncertainty δV directly, taking absolute values (the SPARC convention uses signed velocities for approaching/receding sides).

Four galaxies with fewer than 5 data points were excluded: insufficient degrees of freedom preclude meaningful χ^2 comparison. The remaining 171 galaxies span V_{max} from ~ 20 to $\sim 300 \text{ km s}^{-1}$.

2.2. Fitting procedure

Both models were fitted to each galaxy using `scipy.optimize.differential_evolution` with Sobol initialisation, 10^{-8} tolerance, and polished local refinement. Parameter bounds:

Table 1: Parameter bounds for fitting.

Model	Parameter	Lower	Upper
EFC	$v_{\text{flat}} \text{ (km s}^{-1}\text{)}$	10	350
	$r_{\text{turnon}} \text{ (kpc)}$	0.05	50
	sharpness	0.3	6.0
NFW	$V_{200} \text{ (km s}^{-1}\text{)}$	10	400
	$c \text{ (concentration)}$	1	50

2.3. Metrics

For each galaxy, we compute:

$$\chi^2 = \sum_i \left(\frac{V_{\text{obs},i} - V_{\text{model},i}}{\delta V_i} \right)^2 \quad (1)$$

$$\text{AIC} = \chi^2 + 2k \quad (k = 3 \text{ for EFC, } k = 2 \text{ for NFW}) \quad (2)$$

$$\Delta\text{AIC} = \text{AIC}_{\text{NFW}} - \text{AIC}_{\text{EFC}} \quad (3)$$

Positive ΔAIC favours EFC. The AIC penalty ensures EFC must overcome a +2 handicap from its extra parameter.

2.4. Verdict and regime classification

Kill-Test v6 verdicts are assigned by ΔAIC thresholds: **EFC_decisive** ($> +10$), **EFC** ($[+2, +10]$), **tied** ($[-2, +2]$), **LCDM** ($[-10, -2]$), **LCDM_decisive** (< -10). Regime classification follows the SPARC 175 paper [3]: **FLOW** ($\Delta\text{AIC} \geq +10$, $\chi_{\text{redEFC}}^2 < 20$), **LATENT** ($\Delta\text{AIC} \leq -10$ or $\chi_{\text{redEFC}}^2 > 50$), **TRANSITION** (all others).

3. Results

3.1. Verdict distribution

Table 2: Kill-Test verdict distribution across 171 SPARC galaxies.

Verdict	Count	Percent
EFC_decisive	72	42.1%
EFC	31	18.1%
tied	46	26.9%
LCDM	6	3.5%
LCDM_decisive	16	9.4%
EFC total win rate	103	60.2%

Table 3: Regime distribution.

Regime	Count
FLOW	71
TRANSITION	84
LATENT	16

3.2. Aggregate statistics

Table 4: Summary statistics.

Quantity	Value
Median ΔAIC	+6.21 (favours EFC)
Median χ^2_{red} (EFC)	0.44
Median χ^2_{red} (NFW)	1.69
DDO 154 ΔAIC	+125.2 (sanity check ✓)

3.3. Statistical tests

Table 5: Statistical tests for regime dependence and mass bias.

Test	Statistic	p -value	Interpretation
Mann-Whitney	FLOW vs LATENT	≈ 0	Significant separation
Spearman	$\rho(\Delta\text{AIC}, V_{\text{max}})$	0.15	No mass dependence
	$\rho = 0.11$		

3.4. Notable individual galaxies

Table 6: Extreme cases: top EFC winners and LATENT galaxies.

Galaxy	ΔAIC	Verdict	Note
UGC 11914	+4218	EFC_decisive	Strongest EFC case
UGC 05253	+3965	EFC_decisive	
IC 2574	+1448	EFC_decisive	LSB dwarf
DDO 154	+125.2	EFC_decisive	Original probe-2
NGC 5055	negative	LCDM_decisive	Massive spiral
UGC 02953	negative	LCDM_decisive	Complex baryons
UGC 09133	negative	LCDM_decisive	High surface brightness

4. Discussion

4.1. What this result is

This analysis establishes a specific, bounded claim:

There exists a low-dimensional (3-parameter) functional representation of galaxy rotation curves that outperforms the standard NFW halo profile under equal information-theoretic penalties across 60.2% of the SPARC database.

This is a **model compression test**, not a theory test. The EFC phenomenological curve $v(r) = v_{\text{flat}} \sqrt{1 - \exp(-(r/r_{\text{turnon}})^{\text{sharpness}})}$ compresses rotation curve data more efficiently than NFW in the majority of galaxies, as measured by AIC. The result does not, by itself, validate EFC as a gravitational theory, invalidate dark matter, or establish universality of the EFC action.

4.2. What this result is not

Three distinctions must be made explicit:

(a) Not a theory test. The EFC fitting function is motivated by entropy-gradient phenomenology but is not derived from the EFC covariant action via field equations. Connecting the action to galactic dynamics (Level 3 recovery) remains an open problem. Until that derivation exists, this result tests functional form, not gravitational physics.

(b) Not an equal-footing comparison. The NFW profile is used here in isolation: 2 free parameters (V_{200} , c), no baryonic disc, no bulge, no mass-to-light ratio fitting, no adiabatic contraction. Standard practice in the rotation curve literature (e.g. Li et al. 2020; Katz et al. 2017) fits NFW + disc + bulge with 4–5 parameters. Comparing EFC (3 params) against bare NFW (2 params) is therefore comparing a flexible effective curve against an amputated physical model. A stronger test would compare EFC (3 params) against NFW + baryons (4–5 params); if EFC still wins with *fewer* parameters against a *richer* model, the result becomes substantially harder to dismiss.

(c) Not a Bayesian model comparison. All fits use differential evolution (point estimates). There are no posterior distributions, no marginalized confidence contours, and no Bayesian evidence ($\ln \mathcal{Z}$). The AIC values are therefore necessary but not sufficient for definitive model comparison. Full MCMC or nested sampling (PolyChord/dynesty) would reveal parameter degeneracies and provide proper evidence ratios.

4.3. Cherry-picking refuted

The original Kill-Test v6 probe-2 tested 5 galaxies and found 100% EFC success. The extension to 171 galaxies shows a 60.2% EFC win rate with median $\Delta\text{AIC} = +6.21$. The original sample was, if anything, *conservative*: the full dataset contains galaxies with $\Delta\text{AIC} > 4000$ that were not in the original 5.

The absence of mass dependence (Spearman $\rho = 0.11$, $p = 0.15$) confirms that the EFC advantage is not driven by a subpopulation.

4.4. Strength of the signal

The median $\Delta\text{AIC} = +6.21$ falls at the boundary between “moderate” and “strong” evidence on the Burnham & Anderson scale. This is not overwhelming dominance. However, 72 galaxies (42.1%) exceed $\Delta\text{AIC} > +10$ (“very strong”), and the signal is global: the *median* is positive, not just the tail. The Mann-Whitney test confirms that FLOW and LATENT galaxies are drawn from statistically distinct ΔAIC populations ($p \approx 0$).

4.5. LATENT galaxies: an honest accounting

The 16 LATENT galaxies where NFW wins are predominantly massive spirals. The narrative explanation — “complex baryonic structure” — is **post-hoc**, not predicted. This is a limitation.

To upgrade this from narrative to prediction, future work should:

- Pre-classify galaxies by baryonic complexity (bulge fraction, bar strength, warp index) using SPARC metadata
- Predict which galaxies will be LATENT *before* fitting
- Test whether the LATENT set matches the predicted set

Until this is done, the LATENT interpretation remains descriptive, not explanatory.

4.6. What makes this result non-trivial

Despite the caveats above, three aspects resist easy dismissal:

(1) The functional form is restrictive. The EFC curve is not an arbitrary 3-parameter function. It is monotonically increasing, asymptotically flat, and has a specific exponential saturation structure. A generic 3-parameter polynomial would fit better than this; the fact that *this particular form* — motivated by entropy gradients — outperforms NFW is informative.

(2) The advantage is systematic, not random. A random 60/40 split could arise from noise. But the regime classification shows structure: FLOW galaxies (EFC wins strongly) are statistically separable from LATENT galaxies (NFW wins). This is a structured signal, not scatter.

(3) Three parameters suffice where two do not. NFW has *fewer* parameters and still loses. The AIC penalty ensures EFC must overcome a +2 handicap from its extra parameter. That it does so in 60% of cases suggests that rotation curve data contains a degree of freedom that the NFW profile structurally cannot represent.

4.7. Concrete next steps

Three upgrades would convert this from a Figshare data note to a journal-grade result:

1. **NFW + baryons:** Fit NFW + disc + bulge (4–5 params) using SPARC mass decomposition (V_{gas} , V_{disc} , V_{bul}). If EFC (3 params) still wins against a richer model, the result is decisive.

2. **Full posterior:** Replace differential evolution with nested sampling (dynesty/PolyChord) to obtain $\ln \mathcal{Z}$ and marginalized contours for each galaxy.
3. **Predictive LATENT criterion:** Define a baryonic complexity metric from SPARC metadata and pre-register which galaxies should be LATENT before running the fit.

4.8. Regime architecture connection

This result occupies one cell in the EFC sector decomposition:

- L3 (galactic): EFC phenomenological model wins 60.2% — this paper
- L2 (perturbation sector): μ , Σ , E_G predictions testable with Euclid/DESI — see [5]
- L0 (background): No-Go Theorem closes the $\beta \cdot T(a)$ channel analytically — see [4]

The L3 phenomenological success motivates but does not constitute the L3 theoretical recovery (action $\rightarrow$ field equations $\rightarrow$ galactic limit), which remains the deepest open problem in EFC.

5. Reproducibility

All code and data are publicly available:

- **Script:** `sparc175_killtest_universality.py` (self-contained; requires `scipy`, `numpy`)
- **Output:** `sparc175_killtest_results.json` (116 KB, ledger-compatible, per-galaxy verdicts)
- **Data:** SPARC Rotmod database [2] (`sparc_rotation_curves.dat`, 261 KB)
- **Repository:** <https://github.com/supertedai/EFC>

To reproduce:

```
pip install scipy numpy
python sparc175_killtest_universality.py <data_dir> <output_dir>
```

References

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